

Final Report

Interactive System for Macaw Well-being



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Client: "Planète sauvage" Zoological Park

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Ecole Centrale de Nantes

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1 Introduction

The PEP’S project addresses a concrete animal welfare issue within the Planète Sauvage park. Its objective is to provide a web platform to supervise interactive modules deployed in the field, manage audio content, collect interactions reported by embedded devices, and give caretakers and researchers a tool for daily use. The scope of the Information Systems (IS) project covers the web interface, the backend API, data management, and integration with ESP32-type IoT modules.

From the outset, the project was designed as a multi-actor and multi-profile system. The specifications distinguish between administrators, researchers/caretakers, and IoT modules, with constraints for real-time supervision, role management, sound uploading, and maintaining an architecture robust enough for deployment on an external server. The technical objectives also included high requirements for software quality, documentation, Docker containerization, and the ability to maintain efficient display performance.

Throughout the project, the team gradually transformed a local application base into a much more industrialized system: Docker architecture, Nginx reverse proxy, object storage, asynchronous queues, modernized authentication, quality improvements, reinforced documentation, and effective deployment on a VPS. This final report presents this trajectory synthetically, tracing the overall progress of the project and evaluating the level of goal achievement.

2 Context, Challenges, and Scope

The initial need is based on an animal-machine interaction system intended to enrich the environment of macaws, and later extend the approach to other species like dolphins. The physical system detects an interaction, triggers a sound locally, and then sends the data to a web platform for exploitation. The specifications emphasize several strong expectations: monitoring module status, battery, and IP addresses; multi-species management; audio file uploading and management; remote configuration and interaction visualization; role-based access control; and fast dashboard updates.

Functionally, the most structural requirements were:

- Supervision of modules with status, battery, and IP address;
- Multi-species management;
- Uploading and managing audio files;
- Remote configuration and visualization of interactions;
- Role-based access control;
- Rapid dashboard updates.

Technically, the application had to remain maintainable, dockerized, and documented, while maintaining performance consistent with the target scale: a few administrators, a few researchers/caretakers, and a limited but real fleet of ESP32 modules.

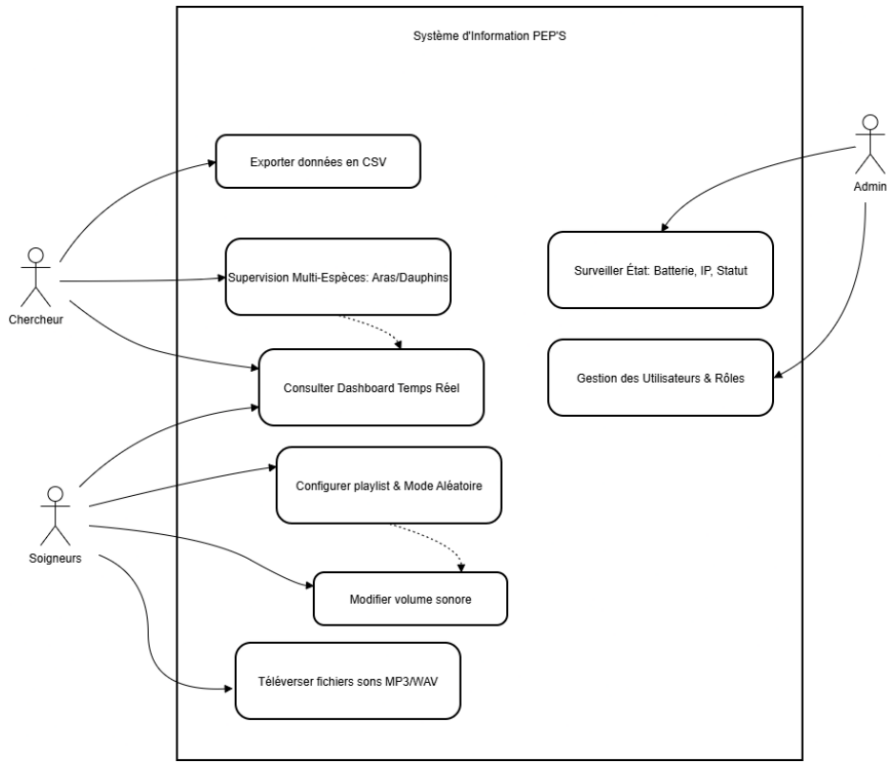


Figure 1: Synthetic use case of the PEP'S system

3 Project Organization and Steering

The start of the project mainly consisted of establishing a common working base, clarifying the framework, and structuring a more rigorous steering method. Initial exchanges with supervisors allowed the specifications to evolve, better define the final scope, explicit user profiles, strengthen the quality dimension, and set more measurable acceptance criteria.

Methodologically, the team then adopted an organization inspired by Agile/Scrum. This transition provided a clearer framework for managing tasks, distributing roles, smoothing validation, and securing deliveries. The project was therefore not limited to an accumulation of developments; it also relied on a rise in maturity regarding work methods.

A major point in the process was the irregular availability of the client. This constraint had a real impact on project agility: some validations were slower than expected, several functional decisions required indirect arbitration, and the team sometimes had to move forward with a higher level of uncertainty than ideal. In this context, Mr. Lechevallier played a structural role by representing the client on several framing points, which secured the functional direction of the application and avoided blockages on architecture, IoT integration, and prioritization.

The Team		
Team Members	Roles	Description
Haytam	Project Owner + Tester	Responsible for the final product vision, managing the backlog, and ensuring product quality through testing.
Ilyas	Scrum Master	Facilitates the Scrum team, helps remove impediments, and ensures Scrum practices are followed.
Dia	Documentation Dev & Client	Manages all project documentation and acts as the primary liaison between the team and the client.
Anas	Front-Back end	Develops and maintains both the user interface (front-end) and the server-side logic and database (back-end).
Clément	Code Quality + SonarCloud	Focuses on maintaining high standards of code quality and utilizing tools like SonarCloud for continuous inspection and improvement.
Santiago	Deployment & Security	Responsible for deploying the application and ensuring its security across all environments.

Figure 2: Project organization and roles in the Agile approach

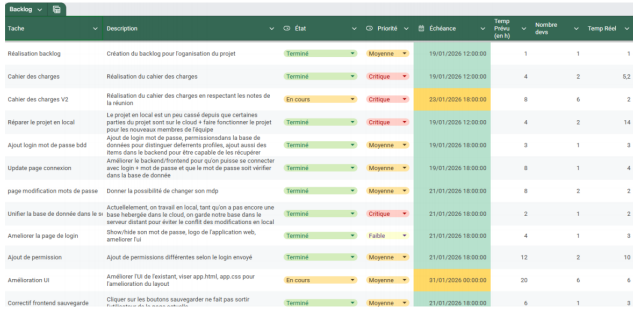
4 Backlog Evolution and Operational Steering

Operational steering occurred in two distinct phases.

Initially, the backlog was kept in a spreadsheet file. This solution offered a very clear advantage: it was free, simple to share, and quick to set up. For a student project with strict budgetary constraints, this approach was consistent at the start. It allowed for structuring initial tasks, adding deadlines, tracking dependencies, and giving an initial overall visibility.

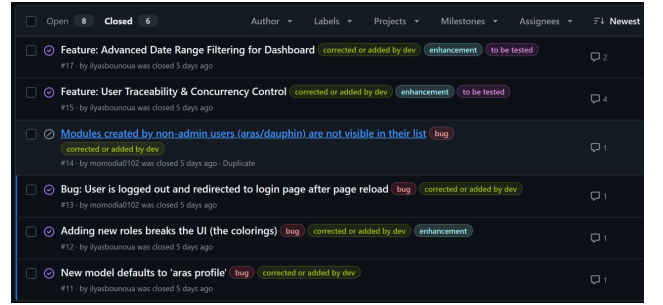
However, this solution showed its limits as the project gained technical depth. Tracking a backlog in an Excel-like file remained less practical when it came to managing development branches, detailed technical tickets, bug fixes, reviews, precise responsibilities, and a direct link to the code repository. This is why this approach was abandoned mid-project in favor of tracking centered on *GitHub Issues*.

This shift constituted true operational progress. Tickets became more technical, better assigned, easier to maintain, and closer to actual implementation. The team thus gained traceability, clarity, and responsiveness, particularly regarding quality, infrastructure, and continuous integration.



Tache	Description	Etat	Priorité	Echéance	Temp. Préf. (h)	Nombre dev	Temp. Réal.
Réalisation backlog	Création du backlog pour l'organisation du projet	Terminé	Moyenne	19/01/2024 12:00:00	1	1	1
Cahier des charges	Réalisation du cahier des charges	Terminé	Critique	19/01/2024 12:00:00	4	2	5,2
Cahier des charges V2	Réalisation du cahier des charges en respectant les notes de la réunion	En cours	Critique	30/01/2024 18:00:00	8	6	2
Réparer le projet en local	Le projet en local est un peu cassé depuis que certaines parties du projet sont sur le cloud → faire fonctionner le projet pour les nouveaux membres de l'équipe	Terminé	Critique	19/01/2024 12:00:00	4	2	14
Ajout login mot de passe bdd	Ajout de login mot de passe, permissions dans la base de données pour distinguer différents profils, ajout aussi des items dans le backend pour être capable de les récupérer	Terminé	Moyenne	19/01/2024 18:00:00	3	1	3
Updater page connexion	Améliorer le backend/frontend pour avoir une page de connexion avec login + mot de passe et que le mot de passe soit vérifié dans la base de données	Terminé	Moyenne	19/01/2024 18:00:00	8	1	4
page modification mots de passe	Donner la possibilité de changer son motp	Terminé	Moyenne	21/01/2024 18:00:00	8	2	2
Unifier la base de données dans le cloud	Actuellement, on travaille en local, tant qu'on a pas encore une base hébergée dans le cloud, on garde notre base dans le serveur distant pour éviter le conflit des modifications en local	Terminé	Critique	21/01/2024 18:00:00	2	1	2
Améliorer la page de login	Shufflez son mot de passe, logo de l'application web, améliorer l'ui	Terminé	Faible	21/01/2024 18:00:00	4	1	3
Ajout de permission	Ajout de permissions différentes selon le login envoyé	Terminé	Moyenne	21/01/2024 18:00:00	12	2	10
Amélioration UI	Améliorer l'UI de l'application, créer app.html, app-ico pour l'installation du logiciel	En cours	Moyenne	31/01/2024 00:00:00	20	8	6
Connectif frontend sauvegarde	Cliquer sur les boutons sauvegarder ne fait pas sortir l'application de la page de connexion	Terminé	Moyenne	21/01/2024 18:00:00	8	1	3

(a) Initial tracking of collaborative work



Open	Closed	Author	Labels	Projects	Milestones	Assignees	Filter	Newest
17	6							
Feature: Advanced Date Range Filtering for Dashboard corrected or added by dev enhancement to be tested 2								
#17: by lyabounous was closed 5 days ago								
Feature: User Traceability & Concurrency Control corrected or added by dev enhancement to be tested 4								
#15: by lyabounous was closed 5 days ago								
Modules created by non-admin users (aras/dauphin) are not visible in their list bug corrected or added by dev 1								
#14: by momodou102 was closed 5 days ago - Duplicate								
Bug: User is logged out and redirected to login page after page reload bug corrected or added by dev 1								
#13: by lyabounous was closed 5 days ago								
Adding new roles breaks the UI (the colorings) bug corrected or added by dev enhancement 1								
#12: by lyabounous was closed 5 days ago								
New model defaults to 'aras profile' bug corrected or added by dev 1								
#11: by lyabounous was closed 5 days ago								

(b) Technical steering via GitHub Issues

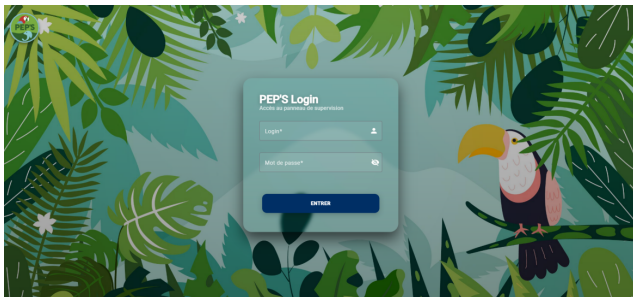
Figure 3: Evolution of backlog steering tools

5 Technical Progress of the Project

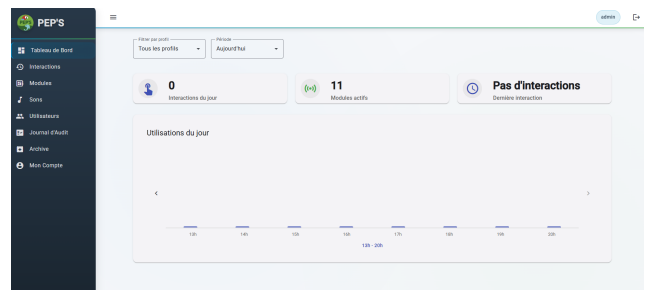
5.1 Establishing the Application Base

The first steps consisted of consolidating the system's foundation. Secure authentication was implemented on the backend with PostgreSQL storage and password encryption. In parallel, the database was centralized on a remote server to facilitate collaborative work and avoid discrepancies between local environments.

On the interface side, an initial redesign of the login page and then the main interface improved the product's readability, general ergonomics, and visual consistency. This phase was important because it allowed for the installation of a reusable, cleaner, and more stable base before increasing the application's complexity.



(a) Login page



(b) Main interface

Figure 4: First stabilized screens of the system

5.2 Functional Structuring and Security

Once this base was established, the team worked on the business functions most useful to the specifications:

- Logging of actions;

- Management of roles and permissions;
- Optimistic locking to avoid write conflicts;
- Support for multi-profile and multi-species;
- Progressive securing of navigation.

The project then passed a significant milestone with the overhaul of authentication to JWT and the use of Angular *Guards* to reinforce access controls. This evolution streamlined the system's operation, improved application security, and made routing cleaner.

5.3 IoT Integration

Integration with the embedded system was one of the project's recurring themes. It was necessary to prepare endpoints compatible with the modules, loosen some security mechanisms on dedicated routes, simulate exchanges, and then validate the feedback of interactions from the field.

A significant difficulty arose when the hardware architecture on the IoT side evolved from a Raspberry Pi-based model to a set of ESP32 boards. Fortunately, this modification did not call into question the general logic of the IS project. The exchange protocol was maintained, and the backend was able to absorb this evolution without major disruption.

Connectivity robustness was then reinforced with a dynamic network configuration mechanism. This addresses a real usage constraint: if the client's local network changes, it becomes possible to reconfigure the boards without heavy maintenance operations. This advancement significantly reduced the operational risk of the system after delivery.

5.4 Architecture Industrialization

One of the most structural changes in the project was the progressive industrialization of the architecture.

The application was first fully dockerized to standardize its execution and prepare for reproducible deployment. An Nginx reverse proxy was then placed as a single entry point to centralize traffic, better control service exposure, and prepare for an operation closer to a production architecture.

The stack was then enriched with:

- PostgreSQL for data persistence;
- MinIO for object storage of audio files;
- Redis, then Redis Streams, for managing asynchronous processing;
- Workers dedicated to processing and storage.

This architecture also allowed for the introduction of backend replica logic and load balancing via Nginx. The project thus moved from an application prototype logic to a logic much closer to a usable system.

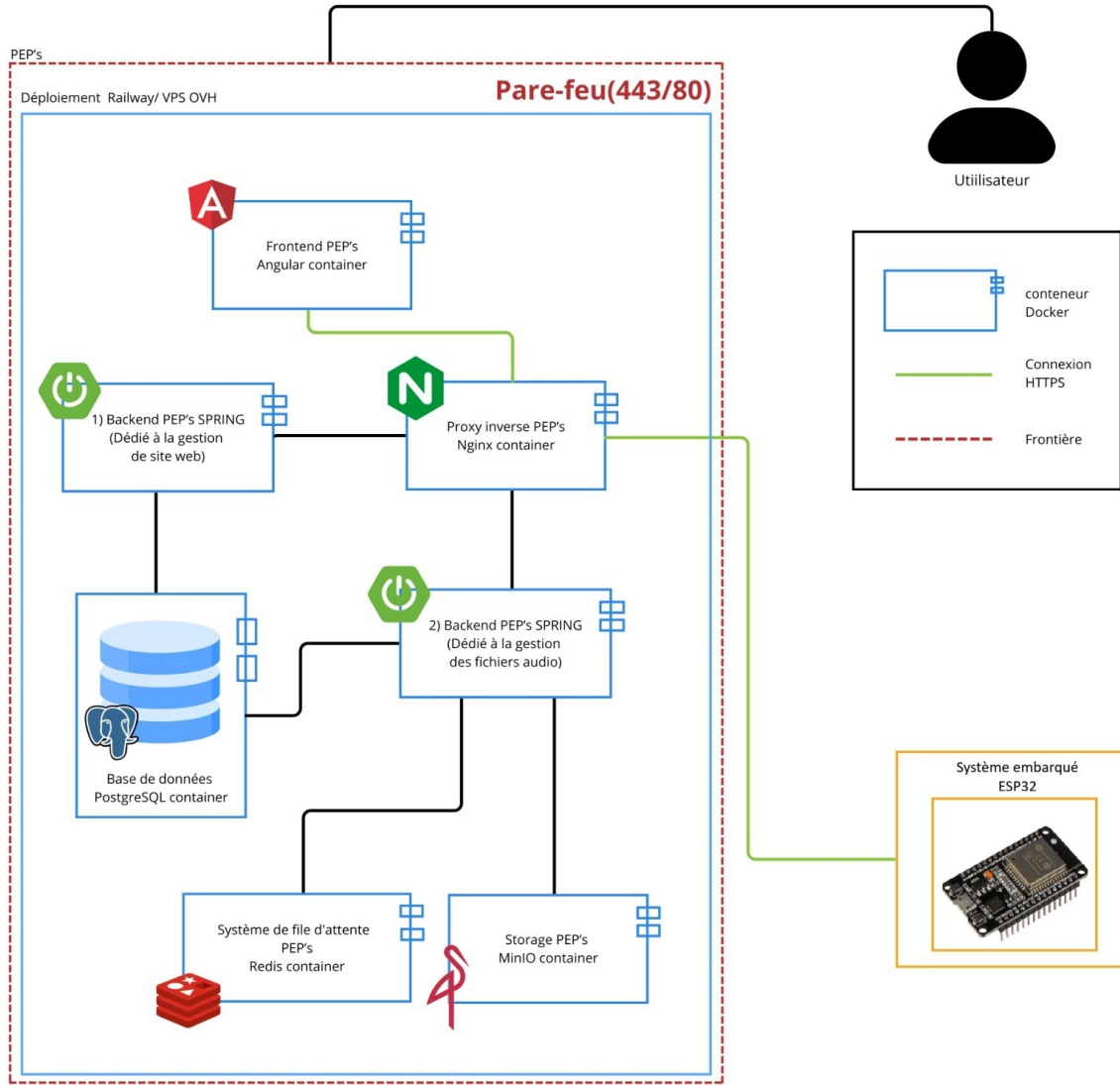


Figure 5: Target industrialized architecture with Nginx, MinIO, and Redis

5.5 Finalized Business Functionalities

At the end of the project, several important functionalities were consolidated:

- Management of audio files and their association with modules;
- Validation of formats and upload constraints;
- Alert system for critical events;
- User profile management page;
- Finalization of the full IoT → backend → database flow;
- Improved responsiveness thanks to the asynchronous pipeline.

Taken together, this forms a consistent application with the expressed need: supervise, configure, historize, and exploit interactions reported from modules deployed in a real environment.

6 Software Quality, Testing, and Documentation

Quality was not treated as a side activity. It accompanied the project throughout its development.

From the first iterations, the team structured a testing logic distinct from development. A validation workflow was established to prevent a member from testing their own developments alone. Simultaneously, functional test inventories were produced to better cover the critical functions of the application.

On both the backend and frontend parts, regular work was also conducted to improve code coverage and address alerts raised by Sonar. The reports show a clear progression in the quality level, particularly on

the frontend part, with a significant gain in coverage. Documentation was progressively enriched, covering software structure, conventions, deployment procedures, and architectural components.

This fundamental work is important in the final assessment, as it explains why the project does not stop at a functional demonstration. It results in a product that is technically easier to take over, more robust, and more credible in a continuity logic.

7 Hosting Choices and Final Deployment

The subject of deployment was studied very early, with a comparison of several approaches and providers. Initially, several options were considered: PaaS, CaaS, VPS, or a more classic deployment. Throughout the project, the choice of an OVH VPS emerged as the most suitable solution.

This choice was motivated by several arguments:

- Good value for money;
- The configuration freedom necessary for a Docker + Nginx + Spring + Angular + Redis + MinIO architecture;
- French location of data centers;
- The impossibility of directly integrating the solution on the client's infrastructure, managed by an external provider.

A major point to remember is that deployment on the VPS was completed as of March 13, 2026. This production launch was made possible because the architecture had been prepared in advance with this perspective: containerization, service separation, environment variables, reverse proxy, and progressive optimization of the infrastructure. In other words, the final deployment was not a last-minute improvisation but the logical conclusion of industrialization work conducted several weeks prior.

The application is currently accessible at a public IP address:

<http://51.75.126.85/login>

This effective availability constitutes a major advancement for the project, even if it remains perfectible as long as no domain name is associated with the instance.

8 Evaluation of Project Success Relative to Specifications

The evaluation of the success rate must remain nuanced. The project clearly achieved a large portion of its structural objectives, but not everything is closed with the same level of maturity.

8.1 Clearly Achieved Elements

The following points can be considered achieved or very largely achieved:

- Operational web architecture;
- Multi-profile management and role-based access control;
- Uploading and management of sounds;
- Integration of IoT/backend exchanges;
- Supervision and data reporting;
- Dockerization of the application;
- Consistent hosting choice and effective deployment on VPS;
- Significantly enriched technical documentation;
- Clear progression in software quality.

8.2 Partially Achieved or To-be-Consolidated Elements

Some subjects remain partial:

- Digital accessibility still in remediation;
- UI finishing still improvable;
- Certain performance and client acceptance criteria not exhaustively demonstrated within the scope of the final report;
- Current absence of a dedicated domain.

8.3 Proposed Success Rate

Using the specifications as the primary reference, and distinguishing between delivered functions, infrastructure blocks, quality, and still partially open topics, the project’s success rate can be estimated at **approximately 85%**.

This value seems defensible for the following reasons:

- Core product functions are present;
- The riskiest technical integration was completed;
- Actual deployment was performed;
- Remaining gaps relate more to finishing, accessibility, certain formal validations, and post-project perspectives than to the functional core.

This rate thus reflects a globally successful, technically serious project, but with normal room for improvement to transition from a very good industrialized prototype to a fully finalized product.

9 Estimation of Project Costs

9.1 Estimation Method

Progress reports do not provide an exhaustive record of hours actually clocked by each member. However, they give several estimated charges for infrastructure tasks, as well as an identified effort for accessibility remediation. To produce a consistent overall estimation, we propose a **reasonable reconstruction** of the project cost, based on:

- Explicit estimations present in the technical backlog;
- The scale of the projects actually completed;
- The overall duration of the project;
- The size of the team (6 students);
- The actually observed load, approximately 50 hours per student.

We thus distinguish:

1. A **core of explicitly visible hours** in the reports;
2. A **consolidated estimation** of the entire project work.

9.2 Hours Explicitly Visible in Reports

Table 1: Explicit charges found in reports

Ref.	Description	Estimated Load
INFRA-001	Implementation of MinIO (service, SDK, controller)	12 to 13 h
INFRA-002	Implementation of Redis as a queue	10 to 12 h
INFRA-003	Configuration of backend replicas	2 to 3 h
TOOL-001	Migration script to MinIO	3 to 5 h
INFRA-004	Nginx optimization (cache, compression)	3 to 5 h
UI-001	Accessibility remediation	8 h
Explicit Total		38 to 46 h

This total, however, only covers a part of the project. Notably missing are:

- Initial design;
- Scoping and updating the specifications;
- Setting up authentication and roles;
- UI overhaul;
- Testing, QA, and SonarCloud;
- Documentation;

- IoT integration;
- Deployment arbitrations;
- Coordination meetings.

9.3 Consolidated Hourly Estimation

Given all the deliverables produced and the load actually observed within the group, a more faithful consolidated estimation of the project is as follows:

Table 2: Consolidated estimation of the project load

Category	Estimated Hours
Scoping, meetings, backlog, steering	40 h
Functional design and architecture	30 h
Backend development	58 h
Frontend development	55 h
IoT integration and network testing	32 h
Infrastructure, Docker, Nginx, VPS, MinIO, Redis	38 h
Quality, testing, SonarCloud, fixes	25 h
Technical documentation and reporting	22 h
Estimated Group Total	300 h

This estimation of **300 hours** seems consistent with:

- The load actually provided by the team;
- The level of system complexity;
- The multiplicity of parallel projects;
- The existence of real industrialization work, and not just a mockup.

9.4 Cost in Hours per Person

With a team of 6 people, the average load per member is:

$$\frac{300}{6} = 50 \text{ hours per person}$$

We can thus retain:

- **Total group cost: 300 h**
- **Average cost per person: 50 h**

9.5 Estimation of Total Cost in Euros

To convert these hours into euros, an assumption must be made. In an academic context, a simple and clear assumption is to use an **average valued cost of 25/h**. This value does not claim to reflect a senior market rate but rather a reasonable valuation of the engineering work performed in a final year project.

The total valued cost then becomes:

$$300 \times 25 = 7,500$$

We therefore retain:

Table 3: Synthesis of costs

Indicator	Value
Total group load	300 h
Average load per person	50 h
Retained hourly rate	25 /h
Estimated total project cost	7,500

Note: With a more conservative valuation at 20/h, the cost would be 6,000; at 30/h, it would be 9,000. For an academic submission, 25/h constitutes a good compromise between credibility and clarity.

10 Difficulties Encountered and Responses Provided

The project faced several difficulties, which influenced its course without calling into question its overall success.

The first difficulty concerned client availability. This situation made some feedback loops more spaced out and reduced the fluidity of ideal Agile steering. To compensate for this, supervisors, particularly Mr. Lechevallier, played an important role in stabilizing the framework and guiding the product on the most sensitive points.

The second difficulty was related to IoT integration. The change in hardware architecture, field constraints, and network connection issues required several adaptations, notably for local network testing and reconnection robustness.

The third difficulty was technical and collaborative: merge conflicts, the need to maintain a stable main branch, the rising complexity of the architecture, and the need for more suitable steering tools than the initial spreadsheet backlog.

Finally, the question of hosting required autonomous arbitration by the team, as the solution could not be integrated directly into the client's servers. This point finally led to a clear, consistent, and controlled choice: the OVH VPS.

11 Perspectives for Improvement

The project constitutes a solid functional base, but several improvements could be considered by a future team to enrich the system and improve its operation:

- **Improved Control of IoT Modules:** Implement a more advanced supervision system to track module status (connectivity, status, battery level) and send automatic notifications when modules become inactive or battery levels are low.
- **Integration of an Artificial Intelligence Layer:** Develop an adaptive mechanism capable of automatically adjusting sound parameters (volume, frequency, type of sound) based on observed animal behavior and interaction history.
- **User Interface Improvements:** Redesign certain screens to improve ergonomics, data visualization, and accessibility for caretakers and researchers.
- **Addition of a Domain Name and HTTPS Securing:** Associate the application with a dedicated domain and configure a full TLS layer to improve the security and readability of the system.
- **Reinforcement of Data Analysis Capabilities:** Develop more advanced dashboards to analyze interactions, compare behaviors between species, and exploit collected data more finely.
- **Improvement of Technical Maintainability:** Continue improving technical documentation, automate more testing, and implement continuous integration pipelines to facilitate the handover and evolution of the project by future teams.

12 Conclusion

The PEP'S project has evolved from a structuring application base into a consistent web platformdockerized, integrated with the IoT system, qualitatively better controlled, and effectively deployed on a VPS. This progression was not achieved solely through the accumulation of functionalities; it was primarily built through successive maturity stages: clarifying requirements, structuring steering, improving the backlog, increasing quality demands, industrializing the architecture, and securing deployment.

Despite a context that was sometimes less favorable than expected, notably due to limited client availability and field-side technical constraints, the team maintained a clear course. The supportive role played by Mr. Lechevallier in framing was decisive in preserving this continuity. Similarly, the transition from a simple backlog to steering via *GitHub Issues*, followed by the choice of a containerized architecture ready for hosting, shows that the project also progressed in its execution.

Regarding the specifications, the result is overall very satisfactory. The core blocks are in place, the product actually exists, the riskiest technical integration was completed, and deployment is effective as of

March 13, 2026. The project can thus be considered a solid success, with an estimated completion rate of approximately 85%, while retaining credible avenues for more ambitious future development.